

Student Take-Home Assignment

Choose at least one of the three challenges. Each takes about 20 to 30 minutes and uses things you already have at home.

All three work the same way, and it is the way real scientists work: make a hypothesis first, then test, then change one thing and test again. Write your hypothesis down before you start. Being wrong is not a problem! In fact, being wrong and figuring out why is the whole point!

DUE FRIDAY, SEPT 25

1. Four Sheets, How Many Books?

You will need about a dozen sheets of paper, tape, and a stack of books. Work on the floor.

- 1 Hypothesize first.** You are going to roll paper into three shapes: a cylinder, a triangle, and a square to try to support a stack of books. Which one do you think will hold the most books? Write down which shape you think will hold the most, write down why, and guess how many books.
- 2 Test the first shape.** Roll or fold four sheets into four identical supports. Stand them up in a square, lay one more sheet across the top, and add books one at a time until it collapses. Count them.
- 3 Test the other two shapes.** Same paper, same height, same tape, same books. Only the shape changes.
- 4 Change exactly one thing.** Take whichever shape won and build it again at half the height. Test it. Stronger or weaker?

Fair test rule: if you change two things at once, you will never know which one mattered. Change one thing, test, then change the next.

Bring: your lab sheet with all four numbers, and one of your paper supports if it survived (or pictures of your supports).

2. Two Squares

Go outside and find two different patches of ground that are near each other but not the same (sunny grass vs. shady grass, a garden bed vs. a sidewalk crack, under a tree vs. out in the open, etc.).

- 1** **Guess first.** Which patch do you think has more living things in it? Write it down and write down why. Then guess how many you will find in each.
- 2** **Mark your squares.** Each one should be about as wide as your arm span. Same size for both.
- 3** **Inspect each one for 10 minutes.** Same amount of time for both. Write down or draw every living thing you find. Look under leaves. Move a rock. Plants count! You do not need to know names. "Tiny orange bug with stripes" is a perfectly good entry.
- 4** **Compare.** Which had more? Was it what you guessed? What was different about the two spots that might explain it?

Fair test rule: same square size, same amount of time, same day. If you spend 20 minutes on one and 5 on the other, the numbers do not mean anything.

Bring: your lab sheet with both lists and both counts, plus where each square was. A photo of each square is a bonus.

3. Instructions Only

This one needs a grown-up who is willing to play along.

- 1** **Build something secretly** out of 8 LEGO bricks (or blocks, or coins, or anything you have several of). Do not let anyone see it.
- 2** **Guess first.** Write instructions using words only. No pictures, no pointing. Then, before you hand them over, guess: out of 8 pieces, how many will your grown-up place correctly?
- 3** **Test it.** Give them the same pieces and your instructions. Leave the room. No talking, no hints, no answering questions. Come back and count how many pieces are in the right place.
- 4** **Change exactly one sentence.** Find the one sentence that caused the most trouble and rewrite just that one. Test again with a different person if you can. Did your score go up?

Bring: your lab sheet, both versions of the sentence you changed, and a photo of your build next to your grown-up's attempt.

Parents: the most useful thing you can do is make your student write the prediction down before they start, and then not rescue them when it turns out wrong. We are not grading results. We are looking for what a student does when the answer surprises them.

Questions? Kate Kremer · katrinakremer@gmail.com · k8kremer.github.io/ddmes-scioly

My Lab Sheet

Name: _____ Grade: _____ Challenge number: _____

1. MY PREDICTION, WRITTEN BEFORE I STARTED

What did you think would happen, and why did you think so?

2. WHAT ACTUALLY HAPPENED

WHAT I TESTED	RESULT

3. THE ONE THING I CHANGED

What did you change, and what happened after you changed it?

4. WAS MY PREDICTION RIGHT?

Circle one: YES NO PARTLY Then tell us what surprised you the most.

Bonus for anyone: crack this
Every letter was shifted forward by 3. A became D, B became E. Shift them back to read it.

GUDJRQIOLHV DUH IDVW IOLHUV

Write your own using a different shift and bring it to stump us.

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